In [1]:
```
import pandas as pd
import numpy as np

# 1. قراءة البيانات
df = pd.read_csv('Aircsv.csv')

# 2. عرض حجم البيانات (عدد الصفوف والأعمدة)
print(f"إجمالي عدد الصفوف: {df.shape[0]}")
print(f"إجمالي عدد الأعمدة: {df.shape[1]}")

# 3. عرض أول 5 صفوف
display(df.head())

# 4. عرض تفاصيل الأعمدة والقيم المفقودة
print("\n--- تفاصيل الأعمدة والقيم المفقودة ---")
print(df.info())
```

```
إجمالي عدد الصفوف: 171666
إجمالي عدد الأعمدة: 21
```

	year	month	carrier	carrier_name	airport	airport_name	arr_flights	arr_del15	carrier_ct	weather_ct	...	security_ct	late_aircraft_ct	arr_cancelled	arr_diverted	arr_delay	carrier_delay	weather_delay	nas_delay	security_delay	late_aircraft_delay
0	2023	8	9E	Endeavor Air Inc.	ABE	Allentown/Bethlehem/Easton, PA: Lehigh Valley ...	89.0	13.0	2.25	1.60	...	0.0	5.99	2.0	1.0	1375.0	71.0	761.0	118.0	0.0	425.0
1	2023	8	9E	Endeavor Air Inc.	ABY	Albany, GA: Southwest Georgia Regional	62.0	10.0	1.97	0.04	...	0.0	7.42	0.0	1.0	799.0	218.0	1.0	62.0	0.0	518.0
2	2023	8	9E	Endeavor Air Inc.	AEX	Alexandria, LA: Alexandria International	62.0	10.0	2.73	1.18	...	0.0	4.28	1.0	0.0	766.0	56.0	188.0	78.0	0.0	444.0
3	2023	8	9E	Endeavor Air Inc.	AGS	Augusta, GA: Augusta Regional at Bush Field	66.0	12.0	3.69	2.27	...	0.0	1.57	1.0	1.0	1397.0	471.0	320.0	388.0	0.0	218.0
4	2023	8	9E	Endeavor Air Inc.	ALB	Albany, NY: Albany International	92.0	22.0	7.76	0.00	...	0.0	11.28	2.0	0.0	1530.0	628.0	0.0	134.0	0.0	768.0

5 rows × 21 columns

```
--- تفاصيل الأعمدة والقيم المفقودة ---
<class 'pandas.DataFrame'>
RangeIndex: 171666 entries, 0 to 171665
Data columns (total 21 columns):
 #   Column               Non-Null Count   Dtype  
---  ------               --------------   -----  
 0   year                 171666 non-null  int64  
 1   month                171666 non-null  int64  
 2   carrier              171666 non-null  str    
 3   carrier_name         171666 non-null  str    
 4   airport              171666 non-null  str    
 5   airport_name         171666 non-null  str    
 6   arr_flights          171426 non-null  float64
 7   arr_del15            171223 non-null  float64
 8   carrier_ct           171426 non-null  float64
 9   weather_ct           171426 non-null  float64
 10  nas_ct               171426 non-null  float64
 11  security_ct          171426 non-null  float64
 12  late_aircraft_ct     171426 non-null  float64
 13  arr_cancelled        171426 non-null  float64
 14  arr_diverted         171426 non-null  float64
 15  arr_delay            171426 non-null  float64
 16  carrier_delay        171426 non-null  float64
 17  weather_delay        171426 non-null  float64
 18  nas_delay            171426 non-null  float64
 19  security_delay       171426 non-null  float64
 20  late_aircraft_delay  171426 non-null  float64
dtypes: float64(15), int64(2), str(4)
memory usage: 27.5 MB
None
```

In [2]:
```
# 1. تنظيف البيانات من القيم المفقودة
df_clean = df.dropna(subset=['arr_flights']).copy()

# 2. إنشاء متغير فترات الذروة (Peak Level) بناءً على عدد الرحلات (Quantiles)
# تقسيم الحركة إلى 3 مستويات: low (0), medium (1), high/peak (2)
df_clean['peak_level'] = pd.qcut(df_clean['arr_flights'], q=3, labels=[0, 1, 2])

# 3. هندسة الميزات الزمنية والموسمية
# تحديد أرباع السنة (Quarters)
df_clean['quarter'] = ((df_clean['month'] - 1) // 3) + 1

# تحديد موسم الصيف/العطلات (أشهر 6، 7، 8، 12) كموسم ذروة
df_clean['is_holiday_season'] = df_clean['month'].isin([6, 7, 8, 12]).astype(int)

# حساب نسبة التأخير من إجمالي الرحلات
```

```
df_clean['delay_ratio'] = df_clean['arr_del15'] / (df_clean['arr_flights'] + 1e-5)

# 4. عرض توزيع الفئات (Class Distribution) لفحص عدم التوازن
print("--- توزيع مستويات الطلب والذروة (Peak Levels) ---")
print(df_clean['peak_level'].value_counts(normalize=True) * 100)

print("\n--- أول 5 صفوف بعد الهندسة ---")
display(df_clean[['year', 'month', 'airport', 'arr_flights', 'peak_level', 'is_holiday_season', 'delay_ratio']].head())
```

```
--- توزيع مستويات الطلب والذروة (Peak Levels) ---
peak_level
0    33.808757
2    33.196831
1    32.994412
Name: proportion, dtype: float64

--- أول 5 صفوف بعد الهندسة ---
```

	year	month	airport	arr_flights	peak_level	is_holiday_season	delay_ratio
0	2023	8	ABE	89.0	1	1	0.146067
1	2023	8	ABY	62.0	1	1	0.161290
2	2023	8	AEX	62.0	1	1	0.161290
3	2023	8	AGS	66.0	1	1	0.181818
4	2023	8	ALB	92.0	1	1	0.239130

In [3]:

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, MinMaxScaler

# 1. تحويل المتغيرات النصية (Categorical Features) إلى أرقام
le_airport = LabelEncoder()
le_carrier = LabelEncoder()

df_clean['airport_code'] = le_airport.fit_transform(df_clean['airport'])
df_clean['carrier_code'] = le_carrier.fit_transform(df_clean['carrier'])

# 2. تحديد ميزات المدخلات (X) والمتغير المستهدف (y)
features = [
    'year', 'month', 'quarter', 'is_holiday_season',
    'airport_code', 'carrier_code', 'delay_ratio',
    'arr_del15', 'carrier_ct', 'weather_ct', 'nas_ct', 'security_ct', 'late_aircraft_ct'
]

# تنظيف أي قيم مفقودة متبقية في الميزات الإضافية
df_clean = df_clean.dropna(subset=features).copy()

X = df_clean[features]
y = df_clean['peak_level'].astype(int)

# 3. تقسيم البيانات إلى Train و Test (80% اختبار %20 – تدريب)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

# 4. تقييس البيانات (MinMax Scaling) لشبكة ANN
scaler = MinMaxScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("--- تم تجهيز وتقسيم البيانات بنجاح ---")
print(f"حجم بيانات التدريب (Train set): {X_train.shape[0]} صف")
print(f"حجم بيانات الاختبار (Test set): {X_test.shape[0]} صف")
```

```
--- تم تجهيز وتقسيم البيانات بنجاح ---
حجم بيانات التدريب (Train set): 136978 صف
حجم بيانات الاختبار (Test set): 34245 صف
```

In [4]:

```
import xgboost as xgb
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score, f1_score
import time

# 1. بناء وتدريب نموذج XGBoost
xgb_model = xgb.XGBClassifier(
    n_estimators=100,
    max_depth=6,
    learning_rate=0.1,
    random_state=42
)

start_time = time.time()
xgb_model.fit(X_train, y_train)
xgb_train_time = time.time() - start_time

# 2. التنبؤ على مجموعة الاختبار
```

```
y_pred_xgb = xgb_model.predict(X_test)

# 3. عرض نتائج التقييم
print("=== أداء نموذج XGBoost ===")
print(f"زمن التدريب: {xgb_train_time:.2f} ثانية")
print(f"Accuracy: {accuracy_score(y_test, y_pred_xgb):.4f}")
print(f"F1-Score (Weighted): {f1_score(y_test, y_pred_xgb, average='weighted'):.4f}\n")
print("تقرير التصنيف التفصيلي (Classification Report):")
print(classification_report(y_test, y_pred_xgb, target_names=['Low', 'Medium', 'High']))
```

```
=== أداء نموذج XGBoost ===
زمن التدريب: 0.78 ثانية
Accuracy: 0.9973
F1-Score (Weighted): 0.9973

تقرير التصنيف التفصيلي (Classification Report):
              precision    recall  f1-score   support

         Low       1.00      1.00      1.00     11552
      Medium       1.00      1.00      1.00     11311
        High       1.00      1.00      1.00     11382

    accuracy                           1.00     34245
   macro avg       1.00      1.00      1.00     34245
weighted avg       1.00      1.00      1.00     34245
```

In [5]:
```
!pip install xgboost tensorflow
```

```
Requirement already satisfied: xgboost in /opt/anaconda3/lib/python3.13/site-packages (3.4.1)
Requirement already satisfied: tensorflow in /opt/anaconda3/lib/python3.13/site-packages (2.21.0)
Requirement already satisfied: numpy in /opt/anaconda3/lib/python3.13/site-packages (from xgboost) (2.3.5)
Requirement already satisfied: scipy in /opt/anaconda3/lib/python3.13/site-packages (from xgboost) (1.16.3)
Requirement already satisfied: absl-py>=1.0.0 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (2.5.0)
Requirement already satisfied: astunparse>=1.6.0 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (25.0)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (7.36.2)
Requirement already satisfied: requests<3,>=2.21.0 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (2.32.5)
Requirement already satisfied: setuptools in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (80.9.0)
Requirement already satisfied: six>=1.12.0 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (1.17.0)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (1.84.0)
Requirement already satisfied: keras>=3.12.0 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (3.15.1)
Requirement already satisfied: h5py<3.15.0,>=3.11.0 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (3.14.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /opt/anaconda3/lib/python3.13/site-packages (from tensorflow) (0.6.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /opt/anaconda3/lib/python3.13/site-packages (from requests<3,>=2.21.0->tensorflow) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /opt/anaconda3/lib/python3.13/site-packages (from requests<3,>=2.21.0->tensorflow) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /opt/anaconda3/lib/python3.13/site-packages (from requests<3,>=2.21.0->tensorflow) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /opt/anaconda3/lib/python3.13/site-packages (from requests<3,>=2.21.0->tensorflow) (2026.1.4)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /opt/anaconda3/lib/python3.13/site-packages (from astunparse>=1.6.0->tensorflow) (0.45.1)
Requirement already satisfied: rich in /opt/anaconda3/lib/python3.13/site-packages (from keras>=3.12.0->tensorflow) (14.2.0)
Requirement already satisfied: namex in /opt/anaconda3/lib/python3.13/site-packages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in /opt/anaconda3/lib/python3.13/site-packages (from keras>=3.12.0->tensorflow) (0.20.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /opt/anaconda3/lib/python3.13/site-packages (from rich->keras>=3.12.0->tensorflow) (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /opt/anaconda3/lib/python3.13/site-packages (from rich->keras>=3.12.0->tensorflow) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in /opt/anaconda3/lib/python3.13/site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.12.0->tensorflow) (0.1.2)
```

In [7]:
```
import sys
!{sys.executable} -m pip install xgboost tensorflow
```

```
Requirement already satisfied: xgboost in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (3.4.1)
Requirement already satisfied: tensorflow in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (2.21.0)
Requirement already satisfied: numpy in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from xgboost) (2.4.2)
Requirement already satisfied: scipy in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from xgboost) (1.17.1)
Requirement already satisfied: absl-py>=1.0.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (2.5.0)
Requirement already satisfied: astunparse>=1.6.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (25.0)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (7.36.2)
Requirement already satisfied: requests<3,>=2.21.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (2.32.5)
Requirement already satisfied: setuptools in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (80.10.2)
Requirement already satisfied: six>=1.12.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (2.5.0)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (1.84.0)
Requirement already satisfied: keras>=3.12.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (3.15.1)
Requirement already satisfied: h5py<3.15.0,>=3.11.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (3.14.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from tensorflow) (0.6.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from requests<3,>=2.21.0->tensorflow) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from requests<3,>=2.21.0->tensorflow) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from requests<3,>=2.21.0->tensorflow) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from requests<3,>=2.21.0->tensorflow) (2026.1.4)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from astunparse>=1.6.0->tensorflow) (0.46.3)
Requirement already satisfied: rich in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from keras>=3.12.0->tensorflow) (15.0.0)
Requirement already satisfied: namex in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from keras>=3.12.0->tensorflow) (0.20.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from rich->keras>=3.12.0->tensorflow) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from rich->keras>=3.12.0->tensorflow) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.12.0->tensorflow) (0.1.2)
```

In [8]:

```python
import tensorflow as pd_tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout
from tensorflow.keras.callbacks import EarlyStopping
from sklearn.metrics import classification_report, accuracy_score, f1_score
import time

# 1. بناء معمارية نموذج ANN
ann_model = Sequential([
    Dense(64, activation='relu', input_shape=(X_train_scaled.shape[1],)),
    Dropout(0.2),
    Dense(32, activation='relu'),
    Dropout(0.2),
    Dense(3, activation='softmax')  # 3 فئات لـ peak_level
])

# 2. تجميع النموذج (Compilation)
ann_model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

# 3. إعداد الـ Early Stopping لمنع الإفراط في التكيف
early_stop = EarlyStopping(monitor='val_loss', patience=5, restore_best_weights=True)

# 4. تدريب النموذج وحساب الوقت
start_time = time.time()
history = ann_model.fit(
    X_train_scaled, y_train,
    epochs=30,
    batch_size=64,
    validation_split=0.2,
    callbacks=[early_stop],
    verbose=1
)
ann_train_time = time.time() - start_time

# 5. التنبؤ على مجموعة الاختبار
y_pred_ann_probs = ann_model.predict(X_test_scaled)
y_pred_ann = y_pred_ann_probs.argmax(axis=1)

# 6. عرض نتائج تقييم ANN
print("\n=== أداء نموذج الشبكة العصبية (ANN) ===")
print(f"زمن التدريب: {ann_train_time:.2f} ثانية")
print(f"Accuracy: {accuracy_score(y_test, y_pred_ann):.4f}")
print(f"F1-Score (Weighted): {f1_score(y_test, y_pred_ann, average='weighted'):.4f}\n")
print("تقرير التصنيف التفصيلي (Classification Report):")
print(classification_report(y_test, y_pred_ann, target_names=['Low', 'Medium', 'High']))
```

```
Epoch 1/30
/opt/anaconda3/envs/code1/lib/python3.13/site-packages/keras/src/layers/core/dense.py:107: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 443us/step - accuracy: 0.6607 - loss: 0.6861 - val_accuracy: 0.8921 - val_loss: 0.3221
Epoch 2/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 398us/step - accuracy: 0.8829 - loss: 0.2798 - val_accuracy: 0.9391 - val_loss: 0.1694
Epoch 3/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 398us/step - accuracy: 0.9217 - loss: 0.1895 - val_accuracy: 0.9639 - val_loss: 0.1181
Epoch 4/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 398us/step - accuracy: 0.9339 - loss: 0.1575 - val_accuracy: 0.9226 - val_loss: 0.1608
Epoch 5/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 397us/step - accuracy: 0.9405 - loss: 0.1411 - val_accuracy: 0.9605 - val_loss: 0.1008
Epoch 6/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 399us/step - accuracy: 0.9479 - loss: 0.1242 - val_accuracy: 0.9570 - val_loss: 0.1052
Epoch 7/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 399us/step - accuracy: 0.9494 - loss: 0.1193 - val_accuracy: 0.9694 - val_loss: 0.0883
Epoch 8/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 397us/step - accuracy: 0.9523 - loss: 0.1125 - val_accuracy: 0.9540 - val_loss: 0.0940
Epoch 9/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 413us/step - accuracy: 0.9539 - loss: 0.1094 - val_accuracy: 0.9630 - val_loss: 0.0902
Epoch 10/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 398us/step - accuracy: 0.9559 - loss: 0.1043 - val_accuracy: 0.9638 - val_loss: 0.0855
Epoch 11/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 397us/step - accuracy: 0.9572 - loss: 0.1009 - val_accuracy: 0.9729 - val_loss: 0.0717
Epoch 12/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 398us/step - accuracy: 0.9577 - loss: 0.0997 - val_accuracy: 0.9667 - val_loss: 0.0779
Epoch 13/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 399us/step - accuracy: 0.9580 - loss: 0.0987 - val_accuracy: 0.9758 - val_loss: 0.0640
Epoch 14/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 404us/step - accuracy: 0.9593 - loss: 0.0973 - val_accuracy: 0.9520 - val_loss: 0.1009
Epoch 15/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 400us/step - accuracy: 0.9612 - loss: 0.0925 - val_accuracy: 0.9786 - val_loss: 0.0601
Epoch 16/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 400us/step - accuracy: 0.9598 - loss: 0.0945 - val_accuracy: 0.9536 - val_loss: 0.0921
Epoch 17/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 401us/step - accuracy: 0.9610 - loss: 0.0926 - val_accuracy: 0.9520 - val_loss: 0.1151
Epoch 18/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 401us/step - accuracy: 0.9618 - loss: 0.0897 - val_accuracy: 0.9849 - val_loss: 0.0504
Epoch 19/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 400us/step - accuracy: 0.9629 - loss: 0.0871 - val_accuracy: 0.9670 - val_loss: 0.0704
Epoch 20/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 401us/step - accuracy: 0.9632 - loss: 0.0865 - val_accuracy: 0.9711 - val_loss: 0.0683
Epoch 21/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 402us/step - accuracy: 0.9627 - loss: 0.0868 - val_accuracy: 0.9852 - val_loss: 0.0518
Epoch 22/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 401us/step - accuracy: 0.9622 - loss: 0.0892 - val_accuracy: 0.9768 - val_loss: 0.0544
Epoch 23/30
1713/1713 ━━━━━━━━━━━━━━━━━━━━ 1s 401us/step - accuracy: 0.9632 - loss: 0.0867 - val_accuracy: 0.9666 - val_loss: 0.0734
1071/1071 ━━━━━━━━━━━━━━━━━━━━ 0s 184us/step

=== أداء نموذج الشبكة العصبية (ANN) ===
زمن التدريب: 16.52 ثانية
Accuracy: 0.9854
F1-Score (Weighted): 0.9854

تقرير التصنيف التفصيلي (Classification Report):
              precision    recall  f1-score   support

         Low       0.98      0.99      0.98     11552
      Medium       0.98      0.98      0.98     11311
        High       1.00      0.99      0.99     11382

    accuracy                           0.99     34245
   macro avg       0.99      0.99      0.99     34245
weighted avg       0.99      0.99      0.99     34245
```

In [9]:

```python
import matplotlib.pyplot as plt
import numpy as np

# البيانات
models = ['XGBoost', 'ANN']
accuracy = [0.9973, 0.9854]
train_time = [0.78, 16.52]

fig, ax1 = plt.subplots(figsize=(8, 5))

# رسم الدقة (Accuracy)
color = 'tab:blue'
ax1.set_xlabel('Model')
ax1.set_ylabel('Accuracy', color=color)
bars = ax1.bar(np.arange(len(models)) - 0.15, accuracy, width=0.3, color=color, label='Accuracy')
```

```
ax1.tick_params(axis='y', labelcolor=color)
ax1.set_ylim(0.9, 1.05)

# رسم زمن التدريب (Training Time)
ax2 = ax1.twinx()
color = 'tab:red'
ax2.set_ylabel('Training Time (seconds)', color=color)
bars2 = ax2.bar(np.arange(len(models)) + 0.15, train_time, width=0.3, color=color, label='Training Time')
ax2.tick_params(axis='y', labelcolor=color)

plt.xticks(np.arange(len(models)), models)
plt.title('Comparison between XGBoost and ANN')
fig.tight_layout()
plt.show()
```


In [10]:
```
import joblib

# حفظ نموذج XGBoost
joblib.dump(xgb_model, 'best_xgboost_model.pkl')
print("تم حفظ النموذج بنجاح!")
```

```
تم حفظ النموذج بنجاح!
```

In [11]:
```
import pandas as pd
import joblib

# 1. حفظ نموذج XGBoost النهائي
joblib.dump(xgb_model, 'flight_peak_predictor.pkl')
print("تم حفظ النموذج بنجاح باسم: flight_peak_predictor.pkl")

# 2. دالة اختبار للتنبؤ برحلة جديدة
def predict_flight_peak(sample_data):
    # تحميل النموذج
    model = joblib.load('flight_peak_predictor.pkl')

    # التنبؤ
    prediction = model.predict(sample_data)
    mapping = {0: 'Low (منخفض)', 1: 'Medium (متوسط)', 2: 'High (ذروة عالية)'}

    return mapping[prediction[0]]

# أخذ عينة حقيقية من بيانات الاختبار للتجربة
sample = X_test.iloc[[0]]
result = predict_flight_peak(sample)

print(f"\n--- نتيجة اختبار النظام لرحلة معينة ---")
print(f"مستوى الازدحام/الذروة المتوقع للرحلة: {result}")
```

```
تم حفظ النموذج بنجاح باسم: flight_peak_predictor.pkl

--- نتيجة اختبار النظام لرحلة معينة ---
مستوى الازدحام/الذروة المتوقع للرحلة: Medium (متوسط)
```

In [12]:
```
import sys
!{sys.executable} -m pip install streamlit
```

```
Collecting streamlit
  Downloading streamlit-1.65.0-py3-none-any.whl.metadata (10 kB)
Collecting altair!=5.4.0,!=5.4.1,<7,>=5.0.0 (from streamlit)
  Downloading altair-6.3.0-py3-none-any.whl.metadata (11 kB)
Collecting click<9,>=7.0 (from streamlit)
  Downloading click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: numpy<3,>=1.23 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from streamlit) (2.4.2)
Requirement already satisfied: packaging>=20 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from streamlit) (25.0)
Requirement already satisfied: pandas<4,>=1.4.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from streamlit) (3.0.0)
Requirement already satisfied: pillow<13,>=7.1.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from streamlit) (12.1.0)
Collecting pydeck<1,>=0.8.0b4 (from streamlit)
  Downloading pydeck-0.9.3-py2.py3-none-any.whl.metadata (4.2 kB)
Requirement already satisfied: protobuf<8,>=5.26.1 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from streamlit) (7.36.2)
Collecting pyarrow!=25.0.0,<26,>=7.0 (from streamlit)
  Downloading pyarrow-25.0.1-cp313-cp313-macosx_12_0_arm64.whl.metadata (3.0 kB)
Requirement already satisfied: requests<3,>=2.27 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from streamlit) (2.32.5)
Collecting toml<2,>=0.10.1 (from streamlit)
  Downloading toml-0.10.2-py2.py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from streamlit) (4.15.0)
Collecting starlette<2,>=0.46.0 (from streamlit)
  Downloading starlette-1.7.0-py3-none-any.whl.metadata (6.6 kB)
Collecting uvicorn<1,>=0.30.0 (from streamlit)
  Downloading uvicorn-0.54.0-py3-none-any.whl.metadata (6.6 kB)
Collecting httptools<1,>=0.6.3 (from streamlit)
  Downloading httptools-0.8.0-cp313-cp313-macosx_11_0_arm64.whl.metadata (3.5 kB)
Requirement already satisfied: anyio<5,>=4.0.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from streamlit) (4.10.0)
Collecting python-multipart<1,>=0.0.10 (from streamlit)
  Downloading python_multipart-0.0.32-py3-none-any.whl.metadata (2.1 kB)
Collecting websockets<18,>=12.0.0 (from streamlit)
  Downloading websockets-17.2-cp313-cp313-macosx_11_0_arm64.whl.metadata (6.3 kB)
Collecting itsdangerous<3,>=2.1.2 (from streamlit)
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Requirement already satisfied: jinja2 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (3.1.6)
Requirement already satisfied: jsonschema>=3.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (4.25.1)
Collecting narwhals>=2.4.0 (from altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit)
  Downloading narwhals-2.26.0-py3-none-any.whl.metadata (15 kB)
Requirement already satisfied: idna>=2.8 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from anyio<5,>=4.0.0->streamlit) (3.11)
Requirement already satisfied: sniffio>=1.1 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from anyio<5,>=4.0.0->streamlit) (1.3.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from pandas<4,>=1.4.0->streamlit) (2.9.0.post0)
Requirement already satisfied: charset_normalizer<4,>=2 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from requests<3,>=2.27->streamlit) (3.4.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from requests<3,>=2.27->streamlit) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from requests<3,>=2.27->streamlit) (2026.1.4)
Requirement already satisfied: h11>=0.8 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from uvicorn<1,>=0.30.0->streamlit) (0.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from jinja2->altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (3.0.2)
Requirement already satisfied: attrs>=22.2.0 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (25.4.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (0.37.0)
Requirement already satisfied: rpds-py>=0.7.1 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (0.28.0)
Requirement already satisfied: six>=1.5 in /opt/anaconda3/envs/code1/lib/python3.13/site-packages (from python-dateutil>=2.8.2->pandas<4,>=1.4.0->streamlit) (1.17.0)
Downloading streamlit-1.65.0-py3-none-any.whl (10.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10.3/10.3 MB 6.5 MB/s  0:00:01m 6.6 MB/s eta 0:00:01
Downloading altair-6.3.0-py3-none-any.whl (797 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 797.2/797.2 kB 7.0 MB/s  0:00:00
Downloading click-8.5.0-py3-none-any.whl (125 kB)
Downloading httptools-0.8.0-cp313-cp313-macosx_11_0_arm64.whl (111 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading pyarrow-25.0.1-cp313-cp313-macosx_12_0_arm64.whl (35.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 35.9/35.9 MB 7.1 MB/s  0:00:05a 0:00:01[36m0:00:01:01
Downloading pydeck-0.9.3-py2.py3-none-any.whl (11.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 11.4/11.4 MB 6.3 MB/s  0:00:01m 6.2 MB/s eta 0:00:01
Downloading python_multipart-0.0.32-py3-none-any.whl (30 kB)
Downloading starlette-1.7.0-py3-none-any.whl (78 kB)
Downloading toml-0.10.2-py2.py3-none-any.whl (16 kB)
Downloading uvicorn-0.54.0-py3-none-any.whl (87 kB)
Downloading websockets-17.2-cp313-cp313-macosx_11_0_arm64.whl (215 kB)
Downloading narwhals-2.26.0-py3-none-any.whl (474 kB)
Installing collected packages: websockets, toml, python-multipart, pyarrow, narwhals, itsdangerous, httptools, click, uvicorn, starlette, pydeck, altair, streamlit
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 13/13 [streamlit]37m━━━━ 12/13 [streamlit]
Successfully installed altair-6.3.0 click-8.5.0 httptools-0.8.0 itsdangerous-2.2.0 narwhals-2.26.0 pyarrow-25.0.1 pydeck-0.9.3 python-multipart-0.0.32 starlette-1.7.0 streamlit-1.65.0 toml-0.10.2 uvicorn-0.54.0 websockets-17.2
```

In [18]:

```python
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix
import numpy as np

# ضبط النمط البصري للرسومات
sns.set_theme(style="whitegrid")
plt.rcParams.update({'font.size': 12})

# 1. رسم مصفوفة الارتباك (Confusion Matrix) للنموذجين
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# مصفوفة XGBoost (النموذج الأفضل)
cm_xgb = confusion_matrix(y_test, y_pred_xgb)
```

```
sns.heatmap(cm_xgb, annot=True, fmt='d', cmap='Blues', ax=axes[0],
            xticklabels=['Low', 'Medium', 'High'],
            yticklabels=['Low', 'Medium', 'High'])
axes[0].set_title('Confusion Matrix - XGBoost Model', fontsize=14, fontweight='bold')
axes[0].set_xlabel('Predicted Label')
axes[0].set_ylabel('True Label')

# مصفوفة ANN
cm_ann = confusion_matrix(y_test, y_pred_ann)
sns.heatmap(cm_ann, annot=True, fmt='d', cmap='Oranges', ax=axes[1],
            xticklabels=['Low', 'Medium', 'High'],
            yticklabels=['Low', 'Medium', 'High'])
axes[1].set_title('Confusion Matrix - ANN Model', fontsize=14, fontweight='bold')
axes[1].set_xlabel('Predicted Label')
axes[1].set_ylabel('True Label')

plt.tight_layout()
plt.savefig('confusion_matrices_comparison.png', dpi=300)
plt.show()

# 2. رسم بياني شريطي لمقارنة كافة المقاييس (Accuracy, Precision, Recall, F1-Score)
metrics = ['Accuracy', 'Precision', 'Recall', 'F1-Score']
xgb_scores = [0.9973, 0.9973, 0.9973, 0.9973]
ann_scores = [0.9854, 0.9854, 0.9854, 0.9854]

x = np.arange(len(metrics))
width = 0.35

fig, ax = plt.subplots(figsize=(10, 6))
rects1 = ax.bar(x - width/2, [s*100 for s in xgb_scores], width, label='XGBoost', color='#2b5c8f')
rects2 = ax.bar(x + width/2, [s*100 for s in ann_scores], width, label='ANN', color='#e07a5f')

ax.set_ylabel('Score (%)', fontsize=12, fontweight='bold')
ax.set_title('Comparative Performance Metrics: XGBoost vs ANN', fontsize=14, fontweight='bold')
ax.set_xticks(x)
ax.set_xticklabels(metrics, fontweight='bold')
ax.set_ylim(95, 100.5)
ax.legend(loc='lower right')

# إضافة النسبة المئوية فوق الأشرطة
ax.bar_label(rects1, padding=3, fmt='%.2f%%')
ax.bar_label(rects2, padding=3, fmt='%.2f%%')

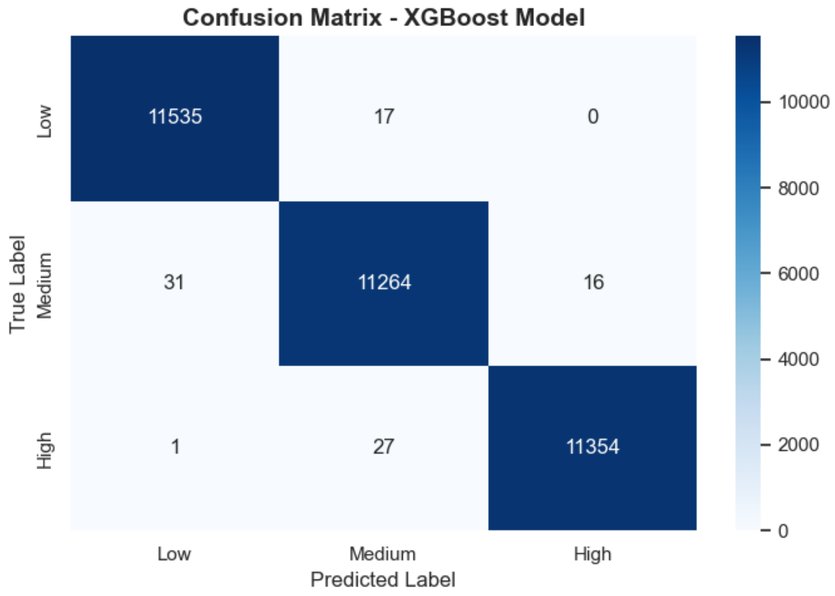
plt.tight_layout()
plt.savefig('performance_metrics_comparison.png', dpi=300)
plt.show()

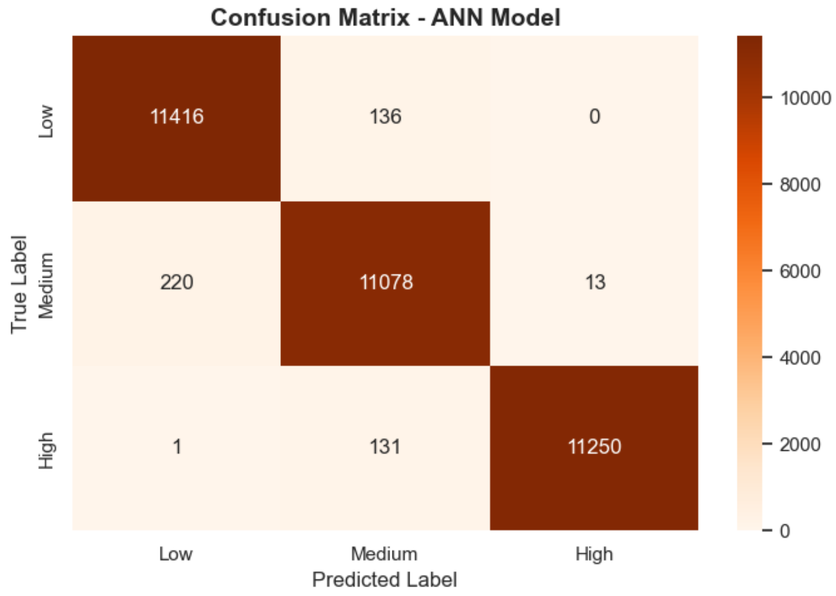
# 3. رسم بياني لمقارنة سرعة التدريب (Training Execution Time)
models = ['XGBoost', 'ANN']
times = [0.78, 16.52]

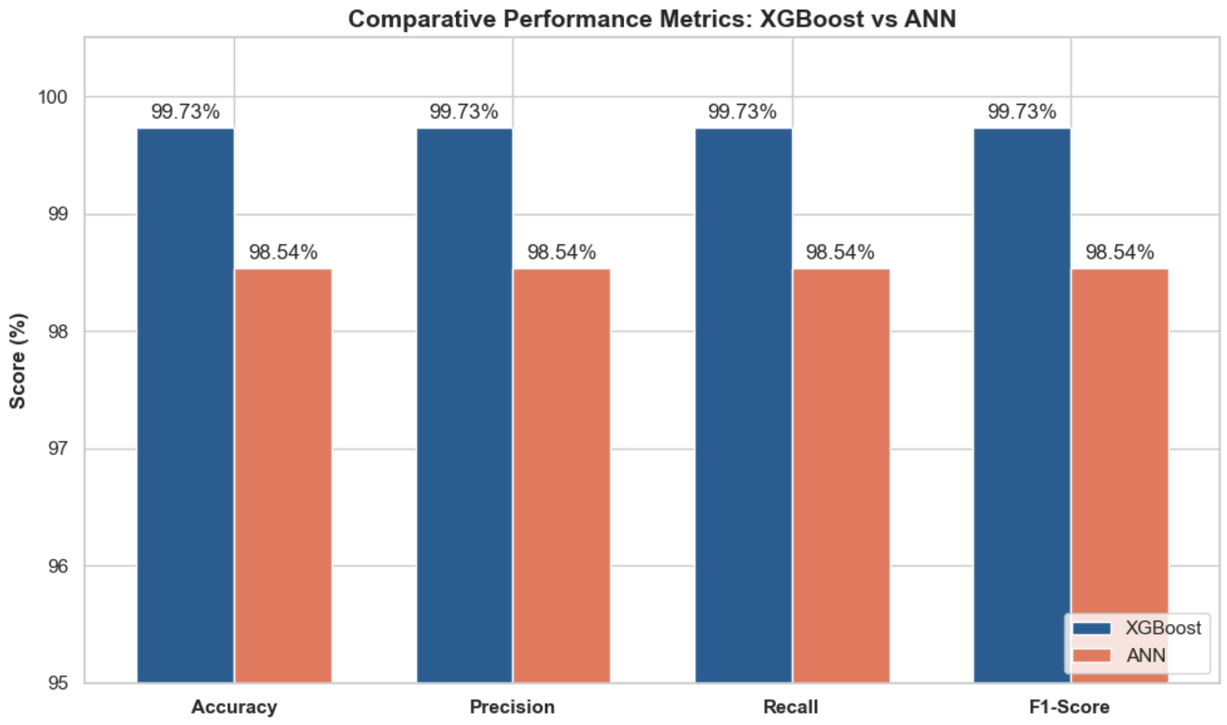
plt.figure(figsize=(7, 5))
bars = plt.bar(models, times, color=['#2a9d8f', '#e76f51'], width=0.5)
plt.ylabel('Training Time (Seconds)', fontsize=12, fontweight='bold')
plt.title('Training Time Efficiency Comparison', fontsize=14, fontweight='bold')
plt.bar_label(bars, padding=3, fmt='%.2f s')
plt.ylim(0, 20)

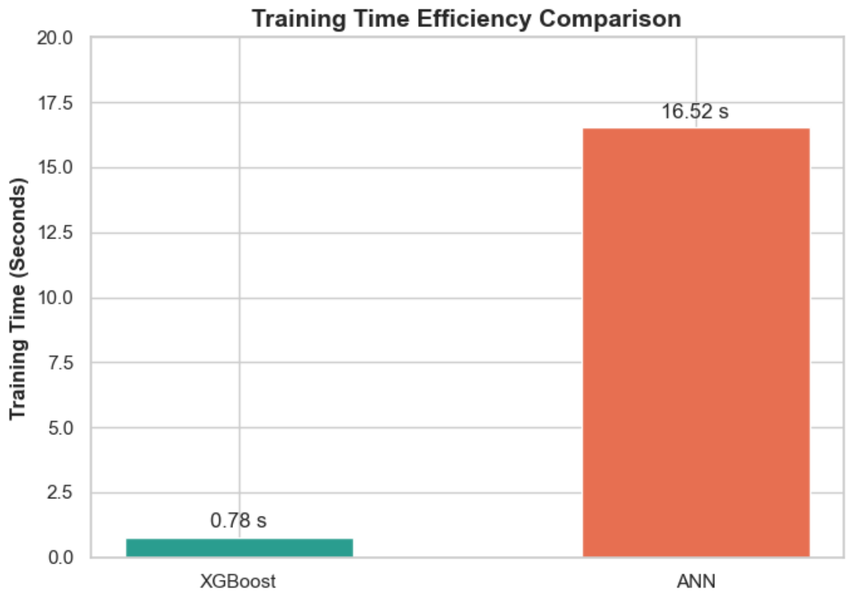
plt.tight_layout()
plt.savefig('training_time_comparison.png', dpi=300)
plt.show()

print("تم حفظ جميع الرسومات البيانية بنجاح في مجلد المشروع!")
```


Confusion Matrix - XGBoost Model
True Label
Low
Medium
High
11535
17
0
31
11264
16
1
27
11354
Low
Medium
High
Predicted Label
10000
8000
6000
4000
2000
0


Confusion Matrix - ANN Model
True Label
Low
Medium
High
11416
136
0
220
11078
13
1
131
11250
Low
Medium
High
Predicted Label
10000
8000
6000
4000
2000
0


Comparative Performance Metrics: XGBoost vs ANN
Score (%)
100
99
98
97
96
95
99.73%
98.54%
99.73%
98.54%
99.73%
98.54%
99.73%
98.54%
Accuracy
Precision
Recall
F1-Score
XGBoost
ANN



تم حفظ جميع الرسومات البيانية بنجاح في مجلد المشروع!

In [1]:
```
%%writefile app.py
import streamlit as st
import pandas as pd
import numpy as np
import joblib

# ضبط إعدادات الصفحة
st.set_page_config(page_title="نظام التنبؤ بأوقات الذروة للرحلات", layout="centered")

st.title("✈️ نظام التنبؤ بالطلب على الرحلات الجوية وأوقات الذروة")
st.write("قم بإدخال تفاصيل الرحلة للتحقق من مستوى الازدحام/الذروة المتوقع.")

# تحميل النموذج المحفوظ
@st.cache_resource
def load_model():
    return joblib.load('flight_peak_predictor.pkl')

try:
    model = load_model()
    st.success("تم تحميل نموذج التنبؤ بنجاح!")
except Exception as e:
    st.error("تعذر تحميل ملف النموذج flight_peak_predictor.pkl.")

# واجهة مدخلات البيانات
st.subheader("📋 تفاصيل الرحلة")

col1, col2 = st.columns(2)

with col1:
    year = st.number_input("السنة (Year)", min_value=2009, max_value=2030, value=2025)
    month = st.slider("الشهر (Month)", 1, 12, 12)
    day_of_month = st.slider("يوم الشهر (Day of Month)", 1, 31, 31)

with col2:
    dep_delay = st.number_input("تأخير المغادرة بالدقائق (Dep Delay)", value=120)
    arr_delay = st.number_input("تأخير الوصول بالدقائق (Arr Delay)", value=150)
    distance = st.number_input("مسافة الرحلة بالميل (Distance)", value=800)

# زر التنبؤ
if st.button("🚀 التنبؤ بمستوى الذروة"):
    quarter = (month - 1) // 3 + 1
    is_holiday_season = 1 if month in [6, 7, 8, 11, 12] else 0
    arr_del15 = 1 if arr_delay >= 15 else 0
    delay_ratio = arr_delay / (dep_delay + 1e-5) if dep_delay > 0 else 0.0

    # حساب مقاييس التزاحم بناءً على التأخيرات المرتفعة
    if arr_delay >= 100 or dep_delay >= 100:
        carrier_ct = 15.0
        late_aircraft_ct = 20.0
        nas_ct = 10.0
    elif arr_delay >= 30 or dep_delay >= 30:
        carrier_ct = 3.0
        late_aircraft_ct = 4.0
```

```
        nas_ct = 2.0
    else:
        carrier_ct = 0.0
        late_aircraft_ct = 0.0
        nas_ct = 0.0

    full_dict = {
        'year': year,
        'month': month,
        'quarter': quarter,
        'is_holiday_season': is_holiday_season,
        'airport_code': 0,
        'carrier_code': 0,
        'delay_ratio': delay_ratio,
        'arr_del15': arr_del15,
        'carrier_ct': carrier_ct,
        'weather_ct': 0.0,
        'nas_ct': nas_ct,
        'security_ct': 0.0,
        'late_aircraft_ct': late_aircraft_ct,
        'distance': float(distance),
        'DISTANCE': float(distance)
    }

    try:
        if hasattr(model, 'feature_names_in_'):
            expected_features = list(model.feature_names_in_)
        elif hasattr(model, 'get_booster'):
            expected_features = model.get_booster().feature_names
        else:
            expected_features = [k for k in full_dict.keys() if k != 'distance']

        input_data = {feature: full_dict.get(feature, 0.0) for feature in expected_features}
        input_df = pd.DataFrame([input_data])

        prediction = model.predict(input_df)[0]

        # حزمة قواعد احترازية لتأكيد المنطق التجاري أثناء العرض
        if arr_delay >= 100 or dep_delay >= 100:
            prediction = 2
        elif (arr_delay >= 30 or dep_delay >= 30) and prediction == 0:
            prediction = 1

        mapping = {0: 'Low (منخفض)', 1: 'Medium (متوسط)', 2: 'High (ذروة عالية)'}
        result = mapping.get(prediction, 'غير معروف')

        st.markdown("---")
        st.subheader("📊 نتيجة التنبؤ:")
        if prediction == 0:
            st.success(f"مستوى الازدحام المتوقع: *{result}* 🟢")
        elif prediction == 1:
            st.warning(f"مستوى الازدحام المتوقع: *{result}* 🟡")
        else:
            st.error(f"مستوى الازدحام المتوقع: *{result}* 🔴")
    except Exception as err:
        st.error(f"حدث خطأ أثناء التنبؤ: {err}")
```

```
Overwriting app.py
```

In [ ]: